Worksheet No. - 1.1

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Semester: 2nd
Date of Performance: 6 jan 2026
Subject Name: Technical training
Subject Code:25CAP-652

Aim/Overview of the practical: To design and implement a sample database system using DDL, DML, and DCL commands, including database creation, data manipulation, schema modification, and role-based access control to ensure data integrity and secure, read-only access for authorized users.

Objective: To gain practical experience in implementing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) operations in a real database environment. This will also include implementing role-based privileges to secure data.

S/W Requirement: Oracle Database Express Edition and pgAdmin

Requirements:

1. **Database Design**
   - Create multiple tables such as **Department**, **Employee**, and **Project**.
   - Define appropriate **PRIMARY KEY** and **FOREIGN KEY** constraints.
   - Enforce **NOT NULL**, **UNIQUE**, and **CHECK** constraints where necessary.
2. **Data Manipulation**
   - Insert sample records into all tables.

- Perform **UPDATE** operations to modify existing records.
- Perform **DELETE** operations while maintaining referential integrity.

3. **Access Control & Security**
   - Create a **role/user** for a reporting staff member.
   - Grant **ONLY SELECT privilege** on required tables to this role/user.
   - Explicitly **REVOKE CREATE privilege** so that the user cannot create any database objects.
   - Ensure the user has **read-only access** to the database.
4. **Schema Modification**
   - Use **ALTER TABLE** to add or modify a column.
   - Drop a table that is no longer required using **DROP TABLE**.

Procedure/Code:

1

```
CREATE TABLE Department (
    dpt_id INT PRIMARY KEY,
    dpt_name VARCHAR(30) UNIQUE NOT NULL
);

CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30) NOT NULL,
```

```
    salary INT CHECK (salary > 0),
    dpt_id INT,
    CONSTRAINT fk_dpt FOREIGN KEY (dpt_id) REFERENCES Department(dpt_id)
);

CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(30) NOT NULL,
    dpt_id INT,
    CONSTRAINT fk_project_dpt FOREIGN KEY (dpt_id) REFERENCES Department(dpt_id)
);

-- 2
INSERT INTO Department VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Sales'),
(4, 'Finance');

INSERT INTO Employee VALUES
(101, 'Roshan', 50000, 2),
(102, 'Sanchit', 45000, 1),
(103, 'Riya', 55000, 3),
```

```
(104, 'Swayam', 60000, 4);

INSERT INTO Project VALUES
(201, 'Payroll System', 2),
(202, 'Web Application', 2);

UPDATE Employee
SET salary = 55000
WHERE emp_id = 101;

DELETE FROM Project
WHERE project_id = 202;

--3
SELECT CURRENT_USER;

CREATE ROLE report_staff
WITH LOGIN PASSWORD 'staff1';

SELECT CURRENT_USER;

GRANT SELECT ON Department TO report_staff;
GRANT SELECT ON Employee TO report_staff;
GRANT SELECT ON Project TO report_staff;
```

```
REVOKE CREATE ON SCHEMA public FROM report_staff;

REVOKE INSERT, UPDATE, DELETE ON Department FROM report_staff;
REVOKE INSERT, UPDATE, DELETE ON Employee FROM report_staff;
REVOKE INSERT, UPDATE, DELETE ON Project FROM report_staff;

--4
ALTER TABLE Employee
ALTER COLUMN emp_name TYPE VARCHAR(50);
DROP TABLE Project;
```

Output:

Data Output Messages Notifications

	dpt_id [PK] integer	dpt_name character varying (30)
1	1	HR
2	2	IT
3	3	Sales
4	4	Finance

Data Output Messages Notifications

```
ERROR:  must be owner of table employee

SQL state: 42501
```

Data Output Messages Notifications

	project_id [PK] integer	project_name character varying (30)	dpt_id integer
1	201	Payroll System	2
2	202	Web Application	2

Learning outcomes (What I have learnt):

- Design relational database tables using appropriate data types, PRIMARY KEY, and FOREIGN KEY constraints.
- Apply data integrity rules using NOT NULL, UNIQUE, and CHECK constraints.
- Insert, update, and delete records from tables using SQL DML commands.
- Maintain referential integrity between related tables using FOREIGN KEY constraints.